

Azizollah Taheri

Taheri.az@northeastern.edu | [LinkedIn](#) | [Google Scholar](#) | 612-999-3175 | Boston, MA

EDUCATION

PhD | *Electrical and Computer Engineering at Northeastern University* Expected 2027, Boston, US
MSc | *Electrical and Computer Engineering at Northeastern University* 2025, Boston, US
BSc | *Mechanical Engineering at Sharif University of Technology* 2021, Tehran, Iran

TECHNICAL SKILLS

Programming: Python, PyTorch, JAX, TensorFlow, OpenCV, MATLAB, C, C++, Julia, R
Robotics Control: Motion Planning, Manipulation, Control Theory, Kinematics, Dynamics
Robotics & Simulation: ROS, Gazebo, Webots
AI / ML: LLMs, VLMs, Reinforcement Learning, Machine Learning, Verifiable Machine Learning
Robot Platforms: TurtleBot, Warthog, Scout Mini, Crazyflie, DJI Tello, IRIS, Vicon, OptiTrack
Planning & Logic: Linear Temporal Logic, Signal Temporal Logic, Time Window Temporal Logic, Planning Domain Definition Language
Tools: SolidWorks, SPSS, Linux, Git

WORK EXPERIENCE

Academic Researcher 2022 – Present

- Developed a cooperative UAV–UGV planning algorithm for semantically unknown environments, using CLIP-based Vision–Language Models for detection and enabling coordinated multi-robot decision-making for space-exploration missions 2026, Boston, US
- Designed a frontier-based temporal-logic motion-planning algorithm for unknown environments, eliminating stagnation and boosting mission success rates by up to 65% with formal correctness guarantees 2025, Boston, US
- Developed an LLM-based planning pipeline for decision-making in unknown environments 2025, Boston, US
- Developed a provably correct motion-planning algorithm for partially known environments, improving decision-making and task safety under temporal logic constraints using dynamic programming — U.S. Army Research Laboratory funded 2025, Boston, US
- Developed a time-window temporal-logic planning algorithm under contingency constraints, enabling robust task scheduling with formal guarantees — supported by the U.S. Army Research Laboratory 2024, Boston, US
- Investigated RL frameworks for probabilistic spatio-temporal planning with time-window constraints, emphasizing robustness and constraint satisfaction — supported by DARPA 2023, Minneapolis, US
- Led experimental studies on trajectory generation integrating state-estimation quality with temporal-logic compliance to improve planning reliability — MnDRIVE funded 2023, Minneapolis, US
- Authored a bachelor's thesis analyzing the kinematics and dynamics of a 2-DOF pick-and-place robot and benchmarking it against two alternative configurations using MATLAB. 2021, Tehran, Iran
- Peer Reviewing:** Reviewed papers for *IEEE RAL*, *IEEE IROS*, *IEEE ICRA* and *IEEE CDC*

Teaching Assistant

- Statics, holding office hours, grading homework and exams, University of Minnesota (Fall 2022)
- Aerospace Vehicle Design, mentoring student projects, University of Minnesota (Fall 2022)
- Machine Learning, holding office hours, grading homework and exams, Northeastern University (Spring 2026)

PUBLICATIONS

- A. Taheri and D. Aksaray, "Motion Planning Under Temporal Logic Specifications In Semantically Unknown Environments", American Control Conference (ACC), 2026 ([link](#)) ([Video](#))
- A. Taheri and D. Aksaray, "Temporal-Logic-Aware Frontier-Based Exploration", American Control Conference (ACC), 2026 ([link](#))

- A. Taheri, Y. M. Phalle and D. Aksaray, *“Temporal Logic Motion Planning of a UAV-UGV Team in Semantically Unknown Environments”*, AIAA 2026 SciTech Forum ([Video](#))
- S. B. Yuksel, A. Taheri, Y. Yazıcıoglu, and D. Aksaray, *“Robust temporal logic planning under contingency constraints”*, in 2025 IEEE 21st International Conference on Automation Science and Engineering (CASE), pp. 30–36, IEEE, 2025 ([link](#))
- A.T. Buyukkocak, Y. Hu, A. Taheri, D. Aksaray, D. Gebre-Egziabher, *“State-Estimation-Aware Planning for Autonomous Systems with Temporal Logic Specifications”*, in AIAA SCITECH 2023 Forum, 2023, p. 2665 ([link](#))